

EEE4114F: Digital Signal Processing

Class Test

14 April 2021

SOLUTIONS

Name:

Student number:

Information

- The test is closed-book.
 - This test has *four* questions, totalling 20 marks.
 - Answer *all* the questions.
 - You have 60 minutes.
 - An information sheet is attached.
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1. (5 marks) A system with input $x[n]$ and corresponding output $y[n]$ is described by the difference equation

$$y[n] = nx[n].$$

Determine with motivation whether the system is a) memoryless, b) causal, c) linear, d) time-invariant, and e) stable.

- (a) To calculate the output at say $n = 10$ only requires knowledge of the input at the same instant. Thus the system is memoryless.
- (b) Because the system is memoryless it is causal, since no future input values are required to calculate a current output value.
- (c) The input $x_1[n]$ produces the output $y_1[n] = nx_1[n]$ and the input $x_2[n]$ produces the output $y_2[n] = nx_2[n]$. The output for $x[n] = ax_1[n] + bx_2[n]$ can be found as follows:

$$y[n] = nx[n] = n(ax_1[n] + bx_2[n]) = anx_1[n] + bnx_2[n] = ay_1[n] + by_2[n].$$

Thus the system is linear.

- (d) The input $x_1[n]$ produces the output $y_1[n] = nx_1[n]$. If the system is time invariant then the input $x_1[n - k]$ must produce the output $y_1[n - k] = (n - k)x_1[n - k]$. However, the output for $x[n] = x_1[n - k]$ is

$$y[n] = nx[n] = nx_1[n - k] \neq y_1[n - k]$$

so the system is not time invariant.

- (e) We can find a bounded input that does not produce a bounded output, so the system is not stable. Consider for example the bounded input $x[n] = u[n]$. The output for $n \geq 0$ is $y[n] = n$. For any proposed upper bound B we can choose a finite n such that $y[n]$ exceeds B , so $y[n]$ is not bounded.

2. (5 marks) Determine the inverse DTFT of each of the following:

(a) $Y_1(e^{j\omega}) = 2\pi$.

(b) $Y_2(e^{j\omega}) = \frac{3}{2-e^{j\omega}}$.

(c) $Y_3(e^{j\omega}) = 5e^{-j3\omega}$.

(d) $Y_4(e^{j\omega}) = 6\cos(3\omega)$.

Using tables we can obtain the following inverse transforms:

(a) $y_1[n] = 2\pi\delta[n]$.

(b) Applying time reversal to the pair $a^n u[n] \Longleftrightarrow \frac{1}{1-ae^{-j\omega}}$ gives

$$a^{-n}u[-n] \Longleftrightarrow \frac{1}{1-ae^{j\omega}}.$$

Since $Y_2(e^{j\omega}) = \frac{3}{2} \frac{1}{1-(1/2)e^{j\omega}}$ we have $y_2[n] = 3/2(1/2)^{-n}u[-n]$.

(c) $y_3[n] = 5\delta[n-3]$.

(d) $y_4[n] = 3\delta[n+3] + 3\delta[n-3]$ since $Y_4(e^{j\omega}) = 3e^{j3\omega} + 3e^{-j3\omega}$.

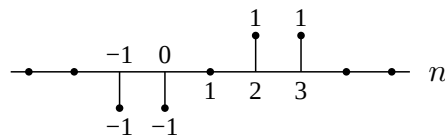
3. (5 marks) Use a time-domain method to determine the convolution of the following two signals and plot the result:

$$x[n] = \delta[n-1] - \delta[n+1] \quad \text{and} \quad v[n] = \delta[n] + \delta[n-1] + \delta[n-2].$$

Since $\delta[n] * v[n] = v[n]$ we know that $\delta[n-1] * v[n] = v[n-1]$ and $\delta[n+1] * v[n] = v[n+1]$. From the other properties of convolution

$$\begin{aligned} x[n] * v[n] &= (\delta[n-1] - \delta[n+1]) * v[n] = \delta[n-1] * v[n] - \delta[n+1] * v[n] \\ &= v[n-1] - v[n+1] \\ &= (\delta[n-1] + \delta[n-2] + \delta[n-3]) - (\delta[n+1] + \delta[n] + \delta[n-1]) \\ &= \delta[n-1] + \delta[n-2] + \delta[n-3] - \delta[n+1] - \delta[n] - \delta[n-1] \\ &= \delta[n-3] + \delta[n-2] - \delta[n] - \delta[n+1]. \end{aligned}$$

The plot follows. This could also be done using graphical methods.



4. (5 marks) Consider the two discrete-time signals $x_1[n] = \delta[n] - \delta[n - 1] + \delta[n - 2]$ and $x_2[n] = 2\delta[n] + 3\delta[n - 1] + 2\delta[n - 2] + \delta[n - 3]$.
- (a) Find expressions for the z-transforms of the sequences $x_1[n]$ and $x_2[n]$.
- (b) Use the previous result to determine and plot the discrete convolution $y[n] = x_1[n] * x_2[n]$.

(a) Since

$$\delta[n] \iff 1 \quad \text{with ROC all } z$$

we can use the frequency-shift property to find the z-transforms of the given signals:

$$X_1(z) = 1 - z^{-1} + z^{-2} \quad \text{and} \quad X_2(z) = 2 + 3z^{-1} + 2z^{-2} + z^{-3}.$$

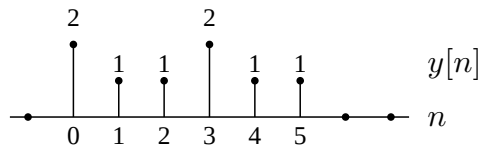
In both cases the ROC is all z .

(b) Using the convolution property for $y[n] = x_1[n] * x_2[n]$ gives

$$\begin{aligned} Y(z) &= X_1(z)X_2(z) = (1 - z^{-1} + z^{-2})(2 + 3z^{-1} + 2z^{-2} + z^{-3}) \\ &= 2 + z^{-1} + z^{-2} + 2z^{-3} + z^{-4} + z^{-5}. \end{aligned}$$

also with ROC all z . Inverting gives the required output:

$$y[n] = 2\delta[n] + \delta[n - 1] + \delta[n - 2] + 2\delta[n - 3] + \delta[n - 4] + \delta[n - 5].$$



Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n + 1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$